

C. STAAD INPUT AND OUTPUT DATA :

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*****
*          STAAD.Pro          *
*      Version  2007    Build 01      *
*      Proprietary Program of          *
*      Research Engineers, Intl.        *
*      Date=    SEP  4, 2008          *
*      Time=    13:45:13              *
*      USER ID: ENGINEERING          *
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1. STAAD SPACE
INPUT FILE: INTERNAL GLASS WALL FRAME3.STD
2. START JOB INFORMATION
3. ENGINEER DATE 13-AUGUST-08
4. JOB NAME GLASS WALL (RASHIDIYA)_INNER
5. JOB CLIENT DMP
6. JOB NO 01
7. JOB REV 0
8. JOB COMMENT PRELIMINARY DESIGN
9. ENGINEER NAME AA
10. END JOB INFORMATION
11. INPUT WIDTH 79
12. UNIT MMS NEWTON
13. JOINT COORDINATES
14. 1 2220 0 0; 2 4440 0 0; 3 6660 0 0; 4 8880 0 0; 5 0 1922 0; 6 13320 1922 0
15. 7 2220 1922 0; 8 4440 1922 0; 9 6660 1922 0; 10 8880 1922 0; 11 0 4422 0
16. 12 13320 4422 0; 13 2220 4422 0; 14 4440 4422 0; 15 6660 4422 0
17. 16 8880 4422 0; 17 0 6922 0; 18 13320 6922 0; 19 2220 6922 0; 20 4440 6922 0
18. 21 6660 6922 0; 22 8880 6922 0; 23 2220 7372 0; 24 4440 7372 0; 25 6660 7372 0
19. 26 8880 7372 0; 27 0.000238419 0 0; 28 0.000238419 7372 0; 29 13320 0 0
20. 30 13320 7372 0; 43 11100 0 0; 44 11100 1922 0; 45 11100 4422 0
21. 46 11100 6922 0; 47 11100 7372 0; 48 2220 1922 -800; 49 4440 1922 -800
22. 50 6660 1922 -800; 51 8880 1922 -800; 52 2220 4422 -800; 53 4440 4422 -800
23. 54 6660 4422 -800; 55 8880 4422 -800; 56 2220 6922 -800; 57 4440 6922 -800
24. 58 6660 6922 -800; 59 8880 6922 -800; 60 11100 1922 -800; 61 11100 4422 -800
25. 62 11100 6922 -800; 73 2220 0 -800; 74 4440 0 -800; 75 6660 0 -800
26. 76 8880 0 -800; 77 2220 7372 -800; 78 4440 7372 -800; 79 6660 7372 -800
27. 80 8880 7372 -800; 81 11100 0 -800; 82 11100 7372 -800
28. MEMBER INCIDENCES
29. 1 1 7; 2 2 8; 3 3 9; 4 4 10; 5 5 7; 6 7 8; 7 8 9; 8 9 10; 9 10 44; 10 7 13
30. 11 8 14; 12 9 15; 13 10 16; 14 11 13; 15 13 14; 16 14 15; 17 15 16; 18 16 45
31. 19 13 19; 20 14 20; 21 15 21; 22 16 22; 23 17 19; 24 19 20; 25 20 21; 26 21 22
32. 27 22 46; 28 19 23; 29 20 24; 30 21 25; 31 22 26; 32 27 5; 33 5 11; 34 11 17
33. 35 17 28; 36 29 6; 37 6 12; 38 12 18; 39 18 30; 40 44 6; 41 45 12; 42 46 18
34. 43 43 44; 44 44 45; 45 45 46; 46 46 47; 47 7 48; 48 8 49; 49 9 50; 50 10 51
35. 51 13 52; 52 14 53; 53 15 54; 54 16 55; 55 19 56; 56 20 57; 57 21 58; 58 22 59
36. 59 44 60; 60 45 61; 61 46 62; 62 73 48; 63 48 52; 64 52 56; 65 56 77; 66 74 49
37. 67 49 53; 68 53 57; 69 57 78; 70 75 50; 71 50 54; 72 54 58; 73 58 79; 74 76 51
38. 75 51 55; 76 55 59; 77 59 80; 78 81 60; 79 60 61; 80 61 62; 81 62 82; 82 45 62
39. 83 61 46; 84 44 61; 85 60 45; 86 43 60; 87 81 44; 88 16 59; 89 55 22; 90 10 55
40. 91 51 16; 92 4 51; 93 76 10; 94 15 58; 95 54 21; 96 9 54; 97 50 15; 98 3 50
41. 99 75 9; 100 14 57; 101 53 20; 102 8 53; 103 49 14; 104 2 49; 105 74 8
42. 106 13 56; 107 52 19; 108 7 52; 109 48 13; 110 1 48; 111 73 7; 112 52 53
43. 113 53 54; 114 54 55; 115 55 61; 116 48 49; 117 49 50; 118 50 51; 119 51 60
44. 120 61 12; 121 16 61; 122 55 45; 123 15 55; 124 54 16; 125 14 54; 126 53 15
45. 127 13 53; 128 52 14; 129 11 52; 130 5 48; 131 7 49; 132 48 8; 133 8 50
46. 134 49 9; 135 9 51; 136 50 10; 137 10 60; 138 51 44; 139 60 6; 140 56 57
47. 141 57 58; 142 58 59; 143 59 62; 144 17 56; 145 19 57; 146 56 20; 147 20 58
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48. 148 57 21; 149 21 59; 150 58 22; 151 22 62; 152 59 46; 153 62 18
49. START GROUP DEFINITION
50. MEMBER
51. _TEN_ROD28DIA 5 TO 9 14 TO 18 23 TO 27 40 TO 42 82 TO 153
52. _50DIA_STRUT 47 TO 61
53. _TEN_ROD38DIA 1 TO 4 10 TO 13 19 TO 22 28 TO 39 43 TO 46 62 TO 81
54. END GROUP DEFINITION
55. DEFINE MATERIAL START
56. ISOTROPIC STEEL
57. E 205000
58. POISSON 0.3
59. DENSITY 7.68195E-005
60. ALPHA 1.2E-005
61. DAMP 0.03
62. END DEFINE MATERIAL
63. MEMBER PROPERTY BRITISH
64. 5 TO 9 14 TO 18 23 TO 27 40 TO 42 82 TO 153 TABLE ST PIPE OD 28 ID 0
65. 47 TO 61 TABLE ST PIPE OD 50 ID 0
66. 1 TO 4 10 TO 13 19 TO 22 28 TO 39 43 TO 46 62 TO 81 TABLE ST PIPE OD 38 ID 0
67. CONSTANTS
68. MATERIAL STEEL ALL
69. SUPPORTS
70. 1 TO 6 11 12 17 18 23 TO 30 43 47 73 TO 82 PINNED
71. MEMBER TENSION
72. 5 TO 9 13 15 TO 27 41 42 44 45 62 63 66 67 70 71 74 82 TO 153
73. LOAD 1 LOADTYPE MASS TITLE SELFWEIGHT
74. SELFWEIGHT Y -1
75. LOAD 2 LOADTYPE DEAD TITLE DEAD LOAD
76. JOINT LOAD
77. 17 18 FY -2092.29
78. 19 TO 22 46 FY -4184.58
79. 11 12 FY -2381.99
80. 13 TO 16 45 FY -4763.98
81. 5 6 FY -2113.75
82. 7 TO 10 44 FY -4227.5
83. 27 29 FY -922.75
84. 1 TO 4 43 FY -1845.51
85. LOAD 3 LOADTYPE WIND TITLE WIND LOAD_POS
86. JOINT LOAD
87. 17 18 FZ -2437.5
88. 19 TO 22 46 FZ -4875
89. 11 12 FZ -2775
90. 13 TO 16 45 FZ -5550
91. 5 6 FZ -2462.5
92. 7 TO 10 44 FZ -4925
93. 27 29 FZ -1075
94. 1 TO 4 43 FZ -2150
95. LOAD 4 LOADTYPE WIND TITLE WIND LOAD_NEG
96. JOINT LOAD
97. 17 18 FZ 2437.5
98. 19 TO 22 46 FZ 4875
99. 11 12 FZ 2775
100. 13 TO 16 45 FZ 5550
101. 5 6 FZ 2462.5
102. 7 TO 10 44 FZ 4925
103. 27 29 FZ 1075
104. 1 TO 4 43 FZ 2150
105. LOAD COMB 5 (SLF + DL + POS_WL)_NOMINAL
106. 1 1.0 2 1.0 3 1.0
107. LOAD COMB 6 (SLF + DL + NEG_WL)_NOMINAL

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108. 1 1.0 2 1.0 4 1.0

109. LOAD COMB 7 (SLF + DL + POS_WL) _FACTORED

110. 1 1.2 2 1.2 3 1.2

111. LOAD COMB 8 (SLF + DL + NEG_WL) _FACTORED

112. 1 1.2 2 1.2 4 1.2

113. PERFORM ANALYSIS

114. PRINT MAX.SECTION DISPL

MEMBER SECTION DISPLACEMENTS

UNIT =INCHES FOR FPS AND CM FOR METRICS/SI SYSTEM

MEMB LOAD GLOBAL X,Y,Z DISPL FROM START TO END JOINTS AT 1/12TH PTS

72	5	-0.0001	-0.0335	-0.4259	-0.0001	-0.0312	-0.4400
		-0.0002	-0.0288	-0.4583	-0.0003	-0.0264	-0.4791
		-0.0003	-0.0241	-0.5005	-0.0004	-0.0217	-0.5207
		-0.0005	-0.0193	-0.5379	-0.0005	-0.0170	-0.5503
		-0.0005	-0.0146	-0.5560	-0.0005	-0.0122	-0.5532
		-0.0004	-0.0099	-0.5402	-0.0003	-0.0075	-0.5150
		-0.0002	-0.0051	-0.4758			

MAX LOCAL DISP = 0.09679 AT 166.67 LOAD 5 L/DISP= 2582

***** END OF SECT DISPL RESULTS *****

115. FOR 28MM DIA TENSION ROD SECTION

116. PRINT MAX.MEMBER FORCES

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- NEWT MMS (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
93	7	76	-10.54	22.70	0.00	0.00	0.00	0.00
		10	119.63	22.70	0.00	0.00	0.00	0.00
	8	76	-33176.66	22.70	0.00	0.00	0.00	0.00
		10	33285.76	22.70	0.00	0.00	0.00	0.00
122	7	55	-9104.13	66.97	0.00	0.00	0.00	0.00
		45	9104.13	66.97	0.00	0.00	0.00	0.00
	8	55	-190.59	66.97	0.00	0.00	0.00	0.00
		45	190.59	66.97	0.00	0.00	0.00	0.00
14	7	11	-9231.30	64.52	0.82	578.86	0.00	22423.68
		13	9231.30	61.49	-0.82	-578.86	-1819.45	-19057.24
	8	11	-164.76	65.79	-4.25	-832.12	0.00	23948.03
		13	164.76	60.22	4.25	832.12	9426.06	-17772.26

117. FOR 50MM DIA STRUT

118. PRINT MAX.MEMBER FORCES

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- NEWT MMS (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
49	7	9	11284.14	53.04	-0.01	0.00	10.98	-15489.06
		50	-11284.14	91.76	0.01	0.00	0.00	0.00
	8	9	7006.86	123.45	-0.04	0.00	33.56	40841.16
		50	-7006.86	21.35	0.04	0.00	0.00	0.00